

QUESTION BANK — Advanced Level (MCQ + Assertion–Reason)

SECTION A — Numbers, Encodings, Logic, Hardware

1. Number Systems & Conversions (Binary, Octal, Decimal, Hex)

MCQs

1. The hexadecimal number $3AF_{16}$ equals what in decimal?
a) 943
b) 879
c) 951
d) 943_{10}
2. Which of the following binary numbers is **NOT** valid for a 4-bit signed 2's complement system?
a) 1000
b) 1111
c) 0111
d) 10100
3. The octal number 725_8 is equal to:
a) 469
b) 421
c) 472
d) 469_{10}
4. The value of $(1011.01)_2$ in decimal is:
a) 11.125
b) 11.25
c) 10.75
d) 11.5

Assertion–Reason

5. **Assertion (A):** Hexadecimal is often used in computing instead of binary.
Reason (R): Every hex digit corresponds to 4 binary bits.
a) A and R true; R explains A
b) A true, R true but unrelated
c) A true, R false
d) A false, R true

2. Binary Arithmetic, 1's & 2's Complement

MCQs

6. The result of $11001_2 + 1011_2$ using binary addition is:
a) 100100
b) 100010
c) 101100
d) 100000
7. The **2's complement** of 00010101_2 is:
a) 11101011
b) 11101010
c) 11101101
d) 00010101
8. In a signed 8-bit 2's complement representation, the decimal value **–60** is:
a) 11000100
b) 10111100
c) 11000110
d) 11000100₂

Assertion–Reason

9. **Assertion (A):** 2's complement arithmetic prevents the existence of “negative zero.”
Reason (R): It allows symmetric positive and negative ranges.
a) A and R true; R explains A
b) A true, R true but unrelated
c) A true, R false
d) A false, R true

3. Floating-Point, Mantissa– Exponent

MCQs

10. If the mantissa is normalized, it means:
a) Mantissa always starts with 1 (for binary)
b) Exponent is always positive
c) Floating number is an integer
d) Both mantissa and exponent are fixed
11. Increasing exponent bits in floating-point format increases:
a) Precision
b) Memory usage only

- c) Range of representable values
- d) Speed of computation

4. Character Encodings (ASCII, Unicode, ISCII)

MCQs

- 12. UTF-8 encoding can represent characters using:
 - a) Fixed 1 byte
 - b) 1–4 bytes
 - c) Only 2 bytes
 - d) Only 4 bytes
- 13. ISCII is mainly used for:
 - a) Chinese
 - b) Indian scripts
 - c) Arabic
 - d) Korean Hangul

Assertion–Reason

- 14. **Assertion (A):** Unicode is backward compatible with ASCII.
Reason (R): ASCII characters map exactly to the first 128 Unicode code points.
 - a) A and R true; R explains A
 - b) A true, R true but unrelated
 - c) A false, R true
 - d) A true, R false

5. Propositional Logic, WFF, Truth Tables

MCQs

- 15. The logical expression $(P \rightarrow Q) \wedge (Q \rightarrow P)$ is equivalent to:
 - a) $P \vee Q$
 - b) $P \leftrightarrow Q$
 - c) $\neg P \vee Q$
 - d) $P \oplus Q$
- 16. Identify the **invalid WFF**:
 - a) $\neg(P \vee Q)$
 - b) $(P \wedge) Q$
 - c) $(P \rightarrow (Q \wedge R))$
 - d) $(\neg P \vee Q)$

Assertion–Reason

- 17. **Assertion (A):** A contradiction is always false.
Reason (R): A contradiction contains opposite statements that cannot be true simultaneously.
 - a) A and R true; R explains A
 - b) A true, R true but unrelated
 - c) A true, R false
 - d) A false, R true

6. Logic Gates & Adders

MCQs

- 18. A **full adder** can be implemented using:
 - a) Two half adders + OR gate
 - b) One half adder only
 - c) XOR gate only
 - d) Decoder
- 19. The output of a **NAND gate** is 0 only when:
 - a) Both inputs are 1
 - b) Both inputs are 0
 - c) Inputs differ
 - d) At least one input is 0

SECTION B — Java Programming Concepts

7. Objects, Classes, JVM

MCQs

- 20. Which among the following is **NOT** a feature of Java?
 - a) Platform independence
 - b) Automatic memory management
 - c) Multiple inheritance using classes
 - d) Security
- 21. Bytecode in Java is executed by:
 - a) Compiler
 - b) JVM
 - c) JRE
 - d) Processor

8. Errors & Exceptions

22. A **runtime exception** is:
- a) Missing semicolon
 - b) Class not found
 - c) ArithmeticException
 - d) Syntax error
23. Finally block executes:
- a) Only if exception occurs
 - b) Only if exception does not occur
 - c) Always executes
 - d) Executes only when JVM decides

9. Types, Casting, Wrapper Classes

24. Which of the following is **not a widening conversion**?
- a) `int` → `long`
 - b) `byte` → `int`
 - c) `float` → `double`
 - d) `double` → `float`
25. The wrapper class for **char** is:
- a) Character
 - b) Char
 - c) C
 - d) Text

10. Variables, Expressions, Precedence

26. In Java, `++x` and `x++` differ because:
- a) Pre-increment updates value before usage
 - b) Post-increment updates value immediately
 - c) Post-increment works only inside loops
 - d) `++x` is slower
27. The highest precedence operator among these is:
- a) `&&`
 - b) `!`
 - c) `+`
 - d) `=`

11. Statements, Scope, Loops

28. A variable declared inside a **for loop initializer** has:
- a) Global scope
 - b) Loop scope
 - c) Method scope
 - d) Class scope
29. The statement **continue** in a loop:
- a) Exits the loop
 - b) Skips current iteration
 - c) Ends the program
 - d) Skips next iteration

12. Methods & Constructors

30. Overloaded methods must differ in:
- a) Return type only
 - b) Parameter list
 - c) Both parameter list & return type
 - d) Method name
31. Which is true for constructors?
- a) They can be overloaded
 - b) They cannot take parameters
 - c) They must return void
 - d) They are called manually

13. Arrays & Strings

32. What will be the output of:
- ```
"abc".concat("xyz").length()
```
- a) 6
  - b) 3
  - c) 5
  - d) 7
33. Which correctly creates a 2D array?
- a) `int[][] a = new int[3][];`
  - b) `int a[][] = new int[][3];`
  - c) `int[][] a = new int[][];`
  - d) `int[3][3] a = new int;`